

Lecture - 10

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

SMALL SAMPLE TESTS

Introduction

For small samples (size < 30); tests proposed for large samples do not hold good as sampling distribution cannot be assumed to be normal for small samples. This led to search of new approaches to deal with small samples. It should be made rational that the methods and theory applicable to small samples can be used for large samples; but the converse is not true.

After hypothesis formulation; choice of test may be sometimes baffling unless specified which test to use. Following suggestions should be kept in mind while choosing test of significance for any hypothesis.

- **Size of sample:** If size of sample is greater than thirty, use any of the applicable large sample tests or Chi-Square test depending upon the applicability.
- **Variance of population:** If the population variance is known and the underlying distribution is normal, t -test should be used. Also, with known

variance; if the distribution is not normal, yet for large sample size, z test can be used. But in case of unknown variance, t-test should be used.

Degree of freedom (ν): The degrees of freedom are the number of independent quantities that can be assigned to any statistical distribution arbitrarily. Suppose we are required to choose 5 numbers whose sum is 30, then we have choice of only 4 numbers. This suggests that a data of size has $(n-1)$ degrees of freedom in general. In case of two restrictions, degrees of freedom will be $(n-2)$.

Student's t-Distribution: Gosset; who wrote under the pen-name 'Student', derived a theoretical distribution known as Student's distribution; which is used to test a hypothesis when the sample size is small and the population variance is not known. As n (size of the sample) increases; distribution tends to normal distribution. It is evident from the graph below that t-distribution has proportionally larger area at its tails than the normal distribution.

If $x_1, x_2, x_3, \dots, x_n$ be a random sample of size $n \leq 30$, from a normal distribution with mean μ and variance σ^2 (not known); also $\bar{x}$ be the sample mean and s standard deviation of the sample

$$\text{Then } t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}, \text{ where } \bar{x} = \frac{\sum x}{n}, S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Degree of freedom is given by $\nu = n - 1$

Statistics is used to test the significance of:

1. Sample mean
2. Difference between two sample means
3. Sample coefficient of correlation
4. Sample Regression Coefficient